fig00_world_dots
figures/static/fig00_world_dots.png
slot 1920 × 1080 px

One Number Can't Tell 8 Billion Stories

Births, deaths and migration in 237 countries and areas — UN estimates 1950–2023, projections to 2100 — and where the UN should look next

111 below replacement · 77 still growing (55 births > deaths · 22 only because more arrive than leave)

Q1 How do births, deaths and migration relate — and how does that differ by region, country and income? P7 · P4 · **Q2** Does one indicator tell the whole story — or only several together? P8 · P5 · **Q3** Do countries share one road at different speeds — and what should the UN watch? P6 · P1

NUMBERS · NOT REAL

DEMO

Which lever moves each country's population: births, deaths or migration?

Population change = births − deaths + net migration

Lever = a component of the population balance, not a policy switch.

Where growth came from, by today's income group

millions of people per decade · births − deaths · net migration

fig01_growth_components_by_income
`figures/animated/fig01_growth_components_by_income.mp4 · figures/static/fig01_growth_components_by_income.png`
slot 721 × 225 px

fig02_lever_map_2023
`figures/static/fig02_lever_map_2023.png`
slot 1037 × 290 px

Today's high-income countries (World Bank FY2027), 2000–2023: net migration ≈ 67 % of population change
56–78 % with other income lists and periods · a group total: it does not mean migration offsets low births country by country

Growing from births	—	Below replacement, still growing from births: — countries, — bn people
Growing only because more arrive than leave	—	– – –
Shrinking: deaths > births	—	– – –
Shrinking: more leave than arrive	—	– – –

Light blue: 22 in 2023 (11–33 depending on the year) · grey: small states (< 1 M) · Map: Natural Earth; boundaries and names do not imply UN endorsement · examples chosen by rule: the three largest countries of each type (≥ 1 M)

ARE **UAE 2023: net migration — per 1,000 people · growth — % a year · fertility — children per woman**
The UN revised the UAE's 2023 population by — % between its 2022 and 2024 editions

UN medium projection, 2050: — M · 95 % range —–– M
95 % range = the UN's band of likely values · — of — countries: the UN 95 % range for 2050 includes both growth and decline

fig03_uae_fan_2050
`figures/static/fig03_uae_fan_2050.png`
slot 560 × 150 px

> **Today's high-income countries: about two in three of growth since 2000 came from net migration; elsewhere, from births. In 22 countries and areas, deaths already outnumber births and net arrivals keep them growing.**

QUESTION 1 · PART 1

NUMBERS · NOT REAL

DEMO

Same road, different speed: child mortality fell first, fertility followed

Same road, different clock: lining countries up on the year fertility fell below 4 children per woman removes — % of the gaps between them

fig04_same_road_registration
`figures/animated/fig04_same_road_registration.mp4 · figures/static/fig04_same_road_registration.png`
slot 774 × 243 px

5 → 3 children per woman: China — years · Egypt — years (fastest and slowest of countries ≥ 10 M that made the trip)

— countries, mostly in sub-Saharan Africa, have not crossed 4 yet and are not shown; — were already below 4 in 1950

Child mortality began a lasting fall first in — of — high-fertility countries (5+ children per woman in 1950; — %) — typically at least — years earlier

child mortality had already fallen by about half (— %)

fig05_mortality_first_event_time
`figures/static/fig05_mortality_first_event_time.png`
slot 986 × 271 px

Fertility began to fall: ≈ — in today's upper-middle-income countries · — lower-middle · — low-income

child mortality was already falling in 1950 in —, so the true gap is probably longer · income groups = World Bank FY2027 (today's groups applied to all years)

Within the same country, halving child mortality goes with — children per woman over the next 20 years — comparing countries, —

fig06_within_vs_across_forest
`figures/static/fig06_within_vs_across_forest.png`
slot 1021 × 183 px

clearest in sub-Saharan Africa (—); not detected in Europe & N. America · each country counts once · association, not cause

Migration: a short memory and almost no neighbourhood

HOW ALIKE? (0 = NO LINK, 1 = IDENTICAL)

same country, one decade to the next: births & deaths ≥ — · migration —

neighbouring countries: fertility — · migration — (— without the Gulf)

Net migration = arrivals − departures, often inferred from census gaps — the least-measured component

QUESTION 1 ANSWERED
P7 · P4

Child mortality fell first and fertility followed, at each country's pace — latest in low-income countries (fertility began to fall ≈ — upper-middle · — lower-middle · — low income). Migration moves on its own — alike among neighbours only in the Gulf — and weighs most in today's high-income countries (≈ two in three of their growth since 2000).

NUMBERS · NOT REAL

DEMO

Does one indicator tell the whole story — or only several together?

What fertility alone captures (adjusted R^2, — countries ≥ 1 M, 2023)

fig07_single_indicator_scorecard

`figures/animated/fig07_single_indicator_scorecard.mp4` · `figures/static/fig07_single_indicator_scorecard.png`

slot 572 × 394 px

R^2 = share of the differences between countries that fertility alone reproduces (0 = none, 1 = all) · same when tested on countries left out · 2050 change is partly built in (UN projections start from fertility) · births − deaths + migration together reach —: true by definition (bookkeeping)

Even the death rate (deaths per 1,000 people) changed meaning: Portugal and Chad share it (— vs —) but are — years apart in life expectancy

Look-alikes = the 5 countries most similar on 12 indicators. Fertility alone finds — % of them · the two summary axes (right): — % · picking at random: — %

LOOK-ALIKES · COL COLOMBIA

by fertility it looks like

— — —

by 12 indicators, like

IRN PER ECU PAN TUN

Colombia: our home country, shown as an illustration; all countries in the repository

TWINS BY RULE · SAME FERTILITY, OPPOSITE GROWTH

ARE UAE · BLR Belarus

same fertility — vs — children per woman → growth ▲ — % vs ▼ **−0.44 %** a year

twins chosen by rule from — pairs (≥ 1 M): same fertility (± 0.05), most different growth

Only — axes stand out from random noise · fertility ≈ axis 1 (r = —) · blind to axis 2 (r = —)

fig09_pca_two_dimensions

`figures/animated/fig09_pca_two_dimensions.mp4` · `figures/static/fig09_pca_two_dimensions.png`

slot 624 × 513 px

axis = one summary of all 12 indicators · r = correlation, from −1 to 1 · axis 1 = — % of the variation (— % depending on which indicators enter)

COMPOSITE INDEX = one score blending 5 indicators (2,000 reasonable ways to build it): weights, not formulas, decide — **— % of countries can move more than 20 places** · a transition-stage index, not a ranking of development or well-being

QUESTION 2 ANSWERED
P8 · P5

No. One indicator tells the slow story (how old a population is), not the fast one (growth, migration). The story emerges when two dimensions, the transition and migration, are read together: same fertility, UAE — % a year vs Belarus −0.44 %.

NUMBERS · NOT REAL

DEMO

One road, four groups — and what to watch on each

How we grouped: each country's 70-year path in fertility, life expectancy, median age and net migration (decade averages) → k-means, which groups similar paths (k = 4) → re-checked with whole-curve methods, 500 random re-runs and neighbours

fig10_trajectory_groups
`figures/animated/fig10_trajectory_groups.mp4 · figures/static/fig10_trajectory_groups.png`
slot 969 × 164 px

— pairs of neighbouring countries share a group (by chance: —) · — countries sit between two groups — a spectrum, not boxes

— % of countries keep their group in at least 4 of 5 re-runs · median age 30 → 40: Fast movers — years · Early movers — years (UN medium projection after 2023)

Closer in life expectancy (typical gap between countries, SD: — → — years), further apart in age (SD — → — years) — and in child deaths: the highest tenth of countries had —× the rate of the lowest tenth in 1950, —× in 2023

fig11_divergence_fan
`figures/static/fig11_divergence_fan.png`
slot 793 × 160 px

life expectancy at 65: SD — → — years · estimates 1950–2023

COUNTRIES WHERE…	ROAD	WATCH / PREPARE	FOR (MANDATE)	WHEN
child deaths fall too slowly: — of — stay above 25 per 1,000 in 2030	—	child survival, newborn care; voluntary family planning; births to adolescent girls	UNICEF · WHO · UNFPA (SDG 3.2, 3.7 · ICPD)	by 2030
fertility is below replacement but the population still grows: — countries, — bn	—	— more years of growth in working and older ages: pensions and long-term care now; no alarm about birth rates	UN DESA · WHO · ILO (Decade of Healthy Ageing · SDG 1.3)	2030–2060
deaths outnumber births: — countries (— %), — by 2050	—	healthy longevity; measure mortality with life expectancy, not the crude death rate	WHO · UN DESA (SDG 3.4)	now–2040
growth comes only from arrivals: 22 countries and areas	—	migrant inclusion; better arrival and departure data; read UN migration projections as scenarios	IOM · UN DESA (GCM obj. 1 & 16 · SDG 10.7)	now
more people leave than arrive: — … · and everywhere: censuses and civil registration	—	decent work, safe return, diaspora links; publish lever types with their sensitivity ranges	IOM · ILO · UNSD (GCM obj. 2 & 19 · SDG 8.8, 17.19)	now / 2030 census round

How we kept it honest · ASA Ethical Guidelines (2022)

Scenarios, not forecasts B.2 · C.4 | **Weakest data, said aloud** B.1 · D.10 | **Patterns, not causes** C.2 · C.3 | **Levers, not verdicts** A.8 · D.6 | **Open & checkable** B.3 · F.4

QUESTION 3 ANSWERED
P6 · P1

Mostly one road at different speeds: three stretches (Early · Fast · Recent movers) plus a migration-led group — and each has something different to watch.

NUMBERS · NOT REAL

References

DEMO

DATA

- United Nations, Department of Economic and Social Affairs, Population Division (2024). World Population Prospects 2024, Online Edition. https://population.un.org/wpp/ — licensed under CC BY 3.0 IGO (https://creativecommons.org/licenses/by/3.0/igo/). Changes: we aggregated, modelled and charted the data; the UN does not endorse this work.
- UN DESA (2024). World Population Prospects 2024: Methodology of the United Nations population estimates and projections. UN DESA/POP/2024/DC/NO.10 · Summary of Results. UN DESA/POP/2024/TR/NO.9 · Release note on 2022 vs 2024 revisions (11 Jul 2024) · WPP 2024 Interim Update (Togo), 19 Jan 2026.
- WPP 2024 Supplement (births by sex, LE60) — prepared by the DataQuest organisers from the UN Data Portal.
- World Bank (2026). Country and Lending Groups, FY2027 (effective 1 July 2026; 2025 GNI per capita, Atlas method). API download 2026-10-22.
- Natural Earth (public domain) map data. Boundaries and names shown do not imply official endorsement or acceptance by the United Nations.

SOFTWARE

- R 4.6.1; tidyverse, sf, rnaturalearth, fixest, survival, fda, cluster, clue, boot, gganimate, av, gifski, tweenr, ggflags, patchwork · reveal.js 6 (MIT) · flag-icons (MIT) · Lato (SIL OFL) · Playwright + Chromium (recording)

METHODS & ETHICS

- American Statistical Association (2022). Ethical Guidelines for Statistical Practice. https://doi.org/10.5281/zenodo.7092386
- Webb, J. W. (1963). The natural and migrational components of population changes in England and Wales. Economic Geography 39(2).
- Coale, A. J. & Watkins, S. C. (eds.) (1986). The Decline of Fertility in Europe (Princeton European Fertility Project: onset of decline).
- Kaplan, E. L. & Meier, P. (1958). JASA 53 · Turnbull, B. W. (1976). JRSS-B 38.
- Ramsay, J. O. & Silverman, B. W. (2005). Functional Data Analysis, 2nd ed. Springer · Kneip, A. & Ramsay, J. O. (2008). JASA 103.
- Mundlak, Y. (1978). Econometrica 46 · Horn, J. L. (1965). Psychometrika 30 (parallel analysis) · Hennig, C. (2007). CSDA 52 (cluster stability).
- Moran, P. A. P. (1950). Biometrika 37 · Efron, B. & Tibshirani, R. (1993). An Introduction to the Bootstrap.
- OECD/JRC (2008). Handbook on Constructing Composite Indicators.
- Okabe, M. & Ito, K. (2008). Color Universal Design · Tol, P. (2021). Colour schemes, SRON (colour palettes).

AI ASSISTANCE & REPRODUCIBILITY

- AI assistance: Claude Code (Anthropic) helped draft code and text; every number was regenerated by the notebook and checked by the team.
- Code, notebook, figures and pre-specified hypotheses (sha256 —): github.com/— (MIT; texts CC BY 4.0).

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